```
        MUL x, x                ; R1:R0 = x × x
        MOV temp, R0            ; temp ←R0
        MUL x, y                ; R1:R0 = x × y
        LSL R0                  ; R0 = 2 × R0
        ADD temp, R0            ; temp = temp + R0
        MUL y, y                ; R1:R0 = y × y
        ADD temp, R0            ; temp = temp + R0

loop:
        RJMP loop               ; loop infinitely
```

Assembly code is then translated to machine-level instructions by the assembler. It will look something like this:

```
Address
Mach. Language Instruction
+00000000:
E015 LDI R18, 0x05
+00000001:
E026 LDI R19, 0x06
+00000002:
9F11 MUL R18, R18
+00000003:
2D00 MOV R17, R0
+00000004:
9F12 MUL R18, R18
+00000005:
0C00 LSL R0
+00000006:
0D00 ADD R17, R0
+00000007:
9F22 MUL R19, R19
+00000008:
0D00 ADD R17, R0
@00000009:

+0000000A:
CFFF RJMP PC-0x0000
```